CIO250 User Product Information

1. Product Information

This document provides important product information regarding the CIO250 IO library.

The information in this User Product Information applies to CIO250 IO library.

2. Delivery Structure

The delivery is organized in the following directory structure:

1. DW_TOWER_tsl18cio250 – the library folder
    a. cdl – cdl netlist for LVS
    b. cds – lef abstract and technology lef
    c. doc – datasheet and application notes
    d. gds – gds2 for 3 top metal options.
    e. synopsys – three corners timing information
    f. verilog – verilog model for all cells.

3. KNOWN PROBLEMS & LIMITATIONS

1. Compilation Error
    a. The compiling error of ‘tsl18cio250_min.lib’: “error: Line 15278, There is a missing timing arc between pins 'PAD' and 'PADR' in the 'pc3d00' cell. (LBDB-23)” should be ignored.
2. Analog IO pad
    a. On page 2-60 of document ds_tsl10cio250.pdf the Analog IO pad ‘PC3D00’ is shown as 1.8V. This PAD is also good for 3.3V, however it was characterized on 1.8V only.

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3. Input Capacitance
    a. The Input Capacitance of pin ‘RENB’ at cell ‘pc3b21eu’ should be 0.12pF. Both the file ‘tsl18cio250_typ.lib’ and the datasheet ‘ds_tsl10cio250.pdf’ on page 2-27 are not updated. The value can be modified manually inside the .lib.
4. Power Rings
    a. The figure on Page 1-3 of “pvcf” cell in ‘ds_tsl10cio250.pdf’ shows cutting of all power rings, while actually the ‘VSS’ power net is connected.
5. Power Pads
    a. According to DR there is a need to have a pair of power supplies per Power domain in order to stand 2k HBM ESD requirements.

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